Effective Postponement of Diabetic Nephropathy With Enalapril in Normotensive Type 2 Diabetic Patients With Microalbuminuria

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OBJECTIVE — This study determines the long-term efficacy of the ACE inhibitor, enalapril, in reducing the progression of microalbuminuria to clinical albuminuria in normotensive patients with type 2 diabetes.

RESEARCH DESIGN AND METHODS — There were 103 normotensive type 2 diabetic patients with persistent albumin excretion rate (AER) 20–200 µg/min and normal renal function followed for 5 years in a prospective randomized single-blind placebo-controlled trial. AER, blood pressure, fasting plasma glucose, and HbA_1 were measured every 3–4 months and glomerular filtration rate (GFR), renal plasma flow (RPF), and urinary urea every 12 months.

RESULTS — In the patients treated with enalapril, AER decreased from 55 ± 33 to 20 ± 59 µg/min (geometric mean ± SD), whereas in the placebo group, AER increased from 53 ± 31 to 85 ± 90 µg/min after 5 years. Within 5 years, 7.7% (4/52) of enalapril-treated subjects and 23.5% (12/51) of placebo-treated subjects progressed to clinical albuminuria defined as AER >200 µg/min and at least 34% above baseline (risk reduction = 66.7%, $P < 0.001$). AER increased at an annual rate of 12.3% (95% CI 9.8–14.9) in the placebo group, while it declined by 16.7% (95% CI −18.3 to −15.2) in the enalapril group ($P < 0.001$). In addition, 8 of the 12 placebo-treated patients had evidence of coronary artery disease. The rest of the parameters remained practically unchanged in the two groups.

CONCLUSIONS — After 5 years of therapy with enalapril, compared with placebo, normotensive subjects with type 2 diabetes experienced significantly less progression of microalbuminuria to clinical albuminuria, reduced AER, and preserved GFR.

Increased urinary albumin excretion rate (AER) in the range of microalbuminuria (20–200 µg/min) predicts overt nephropathy in patients with both type 1 (1,2) and, although not as consistently, type 2 diabetes (3). This clinical syndrome is characterized by persistent albuminuria >200 µg/min, a relentless decline in glomerular filtration rate (GFR), and raised arterial blood pressure (4,5). Because there are about ten patients with type 2 diabetes for every one patient with type 1 diabetes, the absolute number of individuals with type 2 diabetes who have renal failure is probably greater than that for type 1 diabetes (6–8). Therefore, there is a need to evaluate treatments that may delay the progress of nephropathy in type 2 diabetes.

Available data suggest that antihypertensive treatment is the best inhibitor of diabetic nephropathy (9–12). ACE inhibitors have been found more effective than placebo and β-adrenergic blocking agents in hypertensive as well as normotensive diabetes with early and advanced nephropathy (13–23).

We report a relatively long-term 5-year study of the effect of the ACE inhibitor, enalapril, on progression from microalbuminuria to clinical albuminuria in the largest randomized single-blind placebo-controlled trial of normotensive patients with type 2 diabetes and microalbuminuria followed up for 5 years.

RESEARCH DESIGN AND METHODS

— A total of 120 nonobese normotensive patients with type 2 diabetes were recruited at the outpatient clinic of the Endocrine and Metabolic Unit of the Department of Medicine during 1989 and gave informed consent to participate in the study. Because of technical problems (see below), 17 patients were excluded and final analysis was done on 103 patients. Subjects consisting of 60 men and 43 women (49.6 ± 5.2 years old, age at diagnosis of diabetes 40.8 ± 2.6 years, known diabetes duration 9.3 ± 2.4 years) were treated by diet alone (n = 26), diet combined with oral hypoglycemic agents (n = 60), or with insulin (n = 17).

Type 2 diabetes was defined according to World Health Organization criteria: diagnosis after the age of 35 years, having no need for insulin in the 1st year after diagnosis, and having no history of ketosis (24). However, a glucagon test was not performed.

Inclusion criteria were as follows: age ranging between 43 and 55 years; known duration of diabetes <15 years; no history and clinical or laboratory evidence of nondiabetic renal, systemic, cardiac (25), or hepatic diseases; BMI <27 kg/m^2; normal blood pressure values on two consecutive examinations (systolic blood pressure [sBP] ≤140 mmHg, diastolic blood pressure [dBP] ≤90 mmHg); GFR >90 ml/min; and AER of 20–200 µg/min on two con-

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Abbreviations: AER, albumin excretion rate; dBP, diastolic blood pressure; FPG, fasting plasma glucose; GFR, glomerular filtration rate; PAH, sodium aminohippurate; RPF, renal plasma flow; sBP, systolic blood pressure.

secutive visits without evidence of urinary tract infection. However, ultrasonographic assessment for residual urine and diabetic cystopathy was not done.

Seven patients, four taking enalapril and three in the control group, left the study during the 1st year. Four patients on enalapril developed a disturbing cough that persisted for >4 weeks and treatment was discontinued. Six patients (two on enalapril and four in the control group) were lost to follow-up during the second and 3rd year.

The protocol was approved by the ethical committee of the Department of Medicine. After a 3-month pretreatment period, the patients were randomly allocated (26) to receive either 10 mg enalapril daily or placebo (placebo tablets similar but not identical to enalapril) in a single-blind manner. The follow-up period was terminated for each patient exactly 5 years after his or her randomization, and the data were submitted for evaluation.

The patients were followed every 3–4 months, and on these visits, fasting plasma glucose (FPG), HbA_1, serum creatinine, serum and urinary electrolytes, and AER in 24-h urine samples were determined. Blood pressure was measured by a mercury sphygmomanometer with the patients sitting after a 15-min rest; the average of two determinations was recorded. The dBP was determined at Korotkoff phase V. If sBP value of ≥145 mmHg or dBP of ≥95 mmHg were found on two consecutive occasions, treatment with long-acting nifedipine was initiated. Ophthalmoscopy was done yearly by an ophthalmologist, and the presence of diabetic retinopathy was recorded. Renal clearance and plasma renin activity were studied at 0, 6, 12, 24, 36, 48, and 60 months. Urinary urea was also analyzed during the clearance period to exclude the influence of changes in protein intake on renal function.

Table 1—*Clinical and laboratory data at baseline in normotensive type 2 diabetic patients with persistent microalbuminuria*

Variable	Enalapril	Placebo
n	52	51
M/W	30/22	30/21
Mean age (years)	49.8 ± 3.0 (43–58)	50.3 ± 2.1 (45–55)
Age at diagnosis (years)	40.5 ± 2.3 (36–44)	41.0 ± 2.9 (36–46)
Duration of diabetes (years)	9.2 ± 2.4 (5–15)	9.3 ± 2.4 (5–15)
BMI (kg/m²)	22.7 ± 0.8 (3.7–6.9)	23.2 ± 0.7 (21.5–24.4)
FPG (mmol/l)	4.8 ± 7 (3.7–6.9)	4.6 ± 0.9 (4.0–6.7)
HbA_1 (%)	8.0 ± 0.6 (6.8–8.9)	8.1 ± 0.5 (6.5–8.8)
AER (μg/min)	55 ± 33 (22–125)	53 ± 31 (21–132)
GFR (ml · min^{-1} · 1.73 m²)	124 ± 12.2 (95–148)	124 ± 14.6 (92–149)
Mean sBP (mmHg)	132 ± 3.9 (120–138)	132 ± 3.9 (120–138)
Mean dBP (mmHg)	81 ± 3.8 (70–88)	82 ± 3.2 (74–86)
Energy intake (kcal/24 h)	1,694 ± 6.7 (1,500–1,800)	1,690 ± 117.6 (1,500–1,800)
Protein intake (g/24 h)	67 ± 8.6 (46–83)	67 ± 8.8 (46–83)
No. with retinopathy (None/background/proliferative)	25/27/0	23/28/0
Therapy		
Diet alone	14	12
Oral agents	29	31
Insulin	09	08

*Data are *n* or means ± SD (range), except AER, which is given as geometric means ± SD (range).

Laboratory measures

The GFR and renal plasma flow (RPF) were estimated by inulin and sodium aminohippurate (PAH) (27). The tests were performed early in the day in a fasting state. The mean coefficients of variation for GFR were 11.7 and 9.8%, and for RPF, the values were 3.9 and 2.8% in the control and enalapril subjects, respectively ($P > 0.05$). Plasma glucose was measured by the glucose oxidase and HbA_1 by the colorimetric method (28), based on thiobarbituric acid (TBA)-reactive 5-OH-methyl furfural (normal range 7–8%), serum creatinine by a direct colorimetric method (29), electrolytes by flame photometry, and serum albumin by bromocresol green method (30). Cholesterol and triglycerides were measured enzymatically using automated methods (31,32). Lipids were measured in serum and HDL containing supernatant after precipitation of apolipoprotein-B–containing lipoproteins with sodium phosphotungstate and $MgCl_2$ (33). LDL cholesterol was calculated by the Friedewald formula (34). Urinary albumin concentration was measured by radioimmunoassay (Ares Serono Diagnostics, Italy). Protein intake was calculated as 6.25 times total urinary nitrogen excretion (24-h urinary nitrogen plus non-urea nitrogen excretion) estimated to 2 g/day per patient (35,36).

Statistical analysis

All data are expressed as means ± SD and ranges. Values for AER were positively skewed and hence were logarithmically transformed before analysis and are presented as geometric means. To test for adequate randomization, the enalapril and the placebo groups were compared with respect to mean age, duration of diabetes, as well as values of albuminuria, serum creatinine, GFR, HbA_1, sBP, and dBP using pooled variance. Paired and unpaired Student's *t* tests were used for comparisons within and among the groups. The best polynomial fit for the time course of AER, blood pressure, GFR, and RPF was sought with analysis of variance. The best fit was linear in all cases, and the slope of a fitted line to the data was measured by the regression coefficient estimated by least squares. The individual slopes of the two groups were compared by the Student's *t* test. All patient data were included in the analysis, and correlations were sought by stepwise multiple regression analysis, using AER as a dependent variable and sBP, dBP, HbA_1, GFR, and RPF as independent variables. A *P* value of <0.05 was considered significant.

RESULTS — Clinical characteristics of the two groups are reported in Table 1. Age, male/female ratio, known diabetes duration, and other baseline data of the enalapril and the placebo groups showed no significant difference.

The AER in the enalapril group decreased from an initial geometric mean of 55 μg/min to a fifth-year mean of 20 μg/min. In four patients, albuminuria exceeded 200 μg/min and by at least 34% above baseline. In contrast, among the placebo-treated patients, the AER increased from an initial geometric mean of 53 μg/min to a fifth-year geometric mean of 85

μg/min. The AER progressed to clinical albuminuria (>200 μg/min) in 12 patients. The difference between the mean values of daily albuminuria in the two groups became statistically significant after the 1st year; however, the difference in the rate of change in albuminuria from baseline was significant at the end of the 1st year ($P < 0.05$). These data are shown in Fig. 1 and numerically detailed in Table 2. If the development of overt nephropathy or clinical albuminuria (>200 μg/min) is considered important, the risk for developing diabetic nephropathy within 5 years of follow-up was 12/51 (23.5%) in the placebo group compared with 4/52 (7.7%) in the enalapril group. Therefore, enalapril treatment resulted in an absolute risk reduction of 15.8 percentage points for the development of clinical albuminuria over a 5-year period ($P < 0.001$).

The mean sBP and dBP in the enalapril-treated patients remained statistically unchanged (Fig. 2, Table 2). In four patients of this group, sBP ≥145 mmHg or dBP ≥95 mmHg was recorded, and treatment with long-acting nifedipine (10–20 mg twice daily) was added. Of the 51 patients in the placebo group, 12 recorded blood pressure values ≥145/95 mmHg and were treated with long-acting nifedipine. The blood pressure values are shown in Fig. 2 and Table 2.

Renal functions expressed as GFR (Fig. 3) and RPF remained unchanged in both groups during the study period (Table 2). The changes in GFR and RPF produced by enalapril in individual patients were inconsistent and variable.

Metabolic control (Table 2), urinary urea excretion, body weight, serum sodium concentration, and serum lipid concentration did not show significant difference between the two groups during the study period. Serum potassium increased significantly in the enalapril-treated group from 3.69 ± 0.17 to 4.20 ± 0.14 mmol/l and thereafter remained fairly stable at this

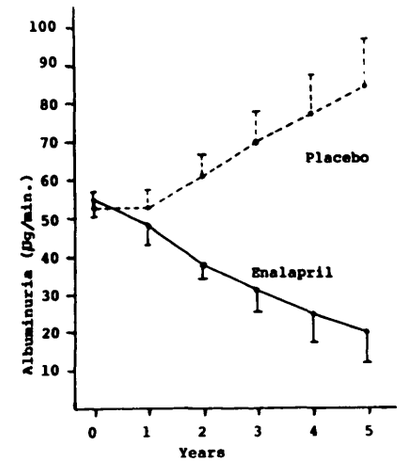

Figure 1—*Albuminuria during the 5-year follow-up in type 2 diabetic subjects treated with enalapril or placebo. Each point on the curve represents the annual average of three to four determinations per patient. Data are geometric means ± SE. Significantly less albuminuria occurred in the enalapril group (P < 0.001) compared with the placebo group over 2–5 years.*

Table 2—*Baseline and annual averages of clinical data and blood chemistry values for the 5-year follow-up in the enalapril and placebo group in type 2 diabetes*

Variable	Treatment	Initial value	1st year	2nd year	3rd year	4th year	5th year
FPG (mmol/l)	Placebo	4.6 ± 0.9 (4.0–6.6)	4.8 ± 0.5 (4.1–6.6)	4.8 ± 0.5 (3.9–6.4)	4.8 ± 0.5 (3.9–6.2)	4.8 ± 0.5 (4.7–6.6)	4.8 ± 0.6 (4.1–6.4)
	Enalapril	4.8 ± 0.7 (3.7–6.9)	4.7 ± 0.6 (3.9–6.6)	4.7 ± 0.6 (3.9–6.6)	4.7 ± 0.6 (3.9–6 4)	4.7 ± 0.6 (3 9–6 5)	4.7 ± 0.6 (4 0–6.4)
HbA_{1c} (%)	Placebo	8.1 ± 0.5 (6.5–8.8)	8.1 ± 0.5 (6.7–8.8)	8.0 ± 8.5 (6.8–8.8)	8.1 ± 0.4 (7.2–8.8)	8.1 ± 0.5 (6.5–8.8)	8.1 ± 0.4 (6.8–8.9)
	Enalapril	8.0 ± 0.6 (6.8–8.9)	8.0 ± 0.6 (6.9–8.8)	8.0 ± 0.6 (6.8–8.8)	8.0 ± 0.6 (7.4–8.8)	8.0 ± 0.6 (7.4–8.8)	8.0 ± 0.5 (7.3–8.8)
sBP (mmHg)	Placebo	132 ± 3 9 (120–138)	133 ± 3.1 (126–138)	133 ± 2.5 (128–138)	132 ± 3.2 (128–138)	135 ± 7.4 (120–150)	134 ± 5 7 (128–148)
	Enalapril	132 ± 3.9 (120–138)	133 ± 3.1 (122–140)	133 ± 2.5 (120–140)	132 ± 3.2 (122–140)	134 ± 7.4 (120–140)	134 ± 5.7 (118–144)
dBP (mmHg)	Placebo	82 ± 3.2 (74–86)	82 ± 3.0 (76–88)	82 ± 2.5 (76–86)	82 ± 2.5 (76–88)	84 ± 6.2 (70–96)	83 ± 5.1 (76–92)
	Enalapril	81 ± 3.8 (70–88)	81 ± 3.6 (72–90)	80 ± 4.8 (70–90)	81 ± 3.4 (72–86)	80 ± 3.6 (72–86)	80 ± 4 6 (70–90)
AER (μg/min)	Placebo	53 ± 31 (21–132)	53 ± 32 (17–129)	61 ± 41 (22–147)	70 ± 60 (23–201)	78 ± 77 (19–242)	85 ± 90 (19–288)
	Enalapril	55 ± 33 (22–125)	48 ± 31 (17–117)	38 ± 30* (13–110)	31 ± 34* (12–176)	25 ± 54* (11–230)	20 ± 59* (07–242)
GFR (ml · min^{-1} · 1.73 m^2 body surface area)	Placebo	124 ± 14.6 (92–149)	122 ± 15.8 (91–148)	121 ± 14.9 (89–146)	120 ± 15.3 (84–145)	119 ± 15.3 (83–146)	119 ± 15.5 (81–145)
	Enalapril	124 ± 12.2 (95–148)	123 ± 12.0 (94–146)	121 ± 12.0 (92–145)	120 ± 12 0 (91–144)	120 ± 12 2 (90–143)	119 ± 12 0 (90–141)
RPF (ml · min^{-1} · 1.73 m^2 body surface area)	Placebo	566 ± 22 3 (535–615)	563 ± 23.4 (523–615)	562 ± 22 9 (513–614)	561 ± 23.8 (509–613)	560 ± 23 7 (508–612)	558 ± 23 6 (506–610)
	Enalapril	580 ± 16.5 (550–635)	584 ± 16.1 (544–630)	585 ± 16.8 (559–618)	585 ± 16.3 (557–629)	585 ± 15.7 (560–624)	585 ± 16.5 (553–642)

Data are means ± SD (range), except AER, which is expressed as geometric means ± SD (range). *$P < 0.05$ for enalapril compared with placebo group.

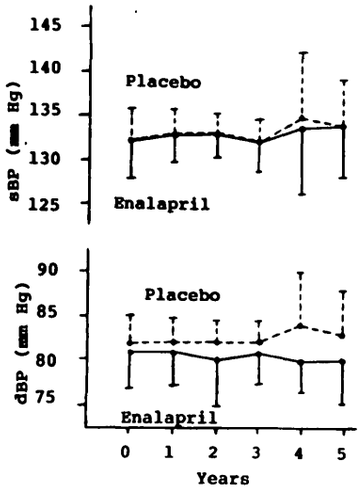

Figure 2—*sBP and dBP values during the 5-year follow-up in type 2 diabetic subjects treated with enalapril or placebo. Data are means ± SD.*

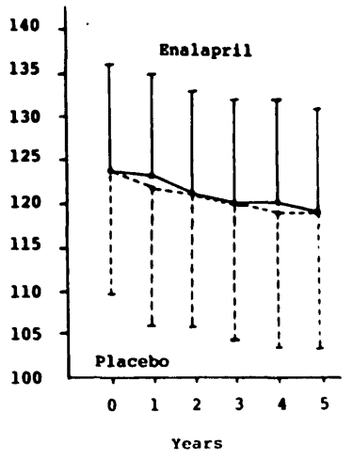

Figure 3—*Time course of GFR during the 5-year follow-up in type 2 diabetic subjects treated with enalapril or placebo. Data are means ± SD.*

increased level for the rest of the observation period. A statistically significant increase in plasma renin activity was observed in the enalapril group from 0.74 ± 0.18 to 2.02 ± 0.59 ng · l^{-1} · s than in the placebo-treated patients ($P < 0.002$), indicating substantial inhibition of ACE activity. Stepwise multiple regression analysis between AER (dependent variable), sBP, dBP, HbA_1, GFR, and RPF (independent variables) showed that reduction in AER was not correlated with the independent variables ($P > 0.05$).

CONCLUSIONS — After 5 years of therapy with the ACE inhibitor, enalapril, normotensive patients with type 2 diabetes had significantly less progression from microalbuminuria to clinical albuminuria than patients treated with placebo ($P < 0.001$). The observed 5-year rate of progression of 7.7% (4/52) in the enalapril-treated group compared favorably with the 23.5% (12/51) rate of progression in the placebo-treated group, a 66.7% reduction after 5 years of therapy ($P < 0.001$). Thus, over a 5-year period, for every six patients treated, progression to clinical albuminuria will be prevented in one patient.

During the 5 years of single-blind study, the mean percentage change in AER from baseline rose in the placebo-treated group and fell significantly in the enalapril-treated group. Further, the annual rate of change revealed an average increase of 12.3% per year with placebo, compared with an average decrease of 16.7% per year with enalapril ($P < 0.001$ between groups).

Similarly, other outcomes were more favorable in the enalapril group because fewer patients needed treatment with nifedipine for hypertension—only 4 on enalapril versus 12 on placebo. In addition, 8 of 12 placebo-treated patients developing clinical albuminuria had electrocardiogram and/or cardiovascular questionnaire evidence of coronary heart disease. The difference in the enalapril- and placebo-treated patients is important in view of the careful control of blood pressure in both groups using nifedipine, when either systolic or diastolic blood pressure increased beyond strictly defined limits. The 5-year changes in blood pressure, however, were not significantly correlated with the changes in AER in either treatment group.

Furthermore, these changes in AER could not be attributed to other variables such as BMI, serum cholesterol, triglycerides, HDL cholesterol, LDL cholesterol, VLDL cholesterol, glycemic control, GFR, and dietary protein intake recognized to cause an impact on albumin excretion. There was no difference in either HbA_1 or urinary urea excretion between the two groups, and there was no correlation between these variables and changes in AER. We measured renal function using inulin and PAH for estimation of GFR and RPF, which is a reliable index of renal function (37).

With enalapril treatment for 5 years, AER was reduced and GFR was preserved, which compared with previously published smaller trials of ACE inhibition (9–23) and also in nondiabetic patients with proteinuria (38,39). The ACE inhibition appears to have a specific renoprotective effect in the stage of incipient diabetic nephropathy, similar to its significant efficacy in preserving kidney function in patients with overt nephropathy, as shown in the Diabetic Nephropathy Collaboration Study (40).

The natural history of nephropathy in type 2 diabetes in India has not been reported, and no data has been published on the long-term intervention and the possibility of true alteration of the natural course of the disease. The ethnic difference in AER pattern between Indian and European patients (41) needs further study and corroboration. This is the first trial to document preservation of renal function with enalapril, initiated during the stage of microalbuminuria in type 2 diabetic subjects. The mechanism of action of ACE inhibition in both the earlier stage of microalbuminuria and in the stage of overt nephropathy may be related to decrease in intraglomerular pressure (42–44) and subsequent preservation of glomeruli, while the decrement in AER after enalapril therapy may result from changes in glomerular hemodynamics. Gansevoort et al. (45) reported a dissociation of the acute renal hemodynamic effects and the antiproteinuric effects of ACE inhibition with enalapril in nondiabetic renal disease.

We found a significant increase in plasma renin activity during enalapril treatment, indicating substantial ACE inhibition and a good compliance; others did not find a significant difference (46,47) perhaps because of anomalous renin secretion described in diabetic patients (48).

Our present study corroborates previous evidence (18) about the renoprotective effect of ACE inhibitors in type 2 diabetes with persistent microalbuminuria. It is possible that ACE inhibition during the stage of microalbuminuria, and possibly even earlier, provides a form of primary prevention and may be particularly helpful in correcting renal hemodynamic alterations induced by hyperglycemia, as reported by Jenkins et al. (49). Further, in cost-benefit analysis, ACE inhibition resulted in substantial cost savings when implemented early (50,51). The efforts to improve glycemic control once persistent microalbuminuria exists do not appear to prevent progression to more advanced stages of albumin excretion, as recently reported (52,53).

The specific renal protection by enalapril and other more potent ACE inhibitors in patients with incipient nephropathy requires further study to

determine the efficacy of this group of drugs in reducing the morbidity and mortality in these patients. After completion of the study, all patients in the control group were switched over to enalapril.

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